

MA2006B - Modern Algebras for Security and Cryptography

Lecture Notes

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1 Group Theory

1.1 Introduction

Group theory is the study of algebraic structures known as groups. The concept of a group is central to abstract algebra and has fundamental applications in cryptography, particularly in public-key cryptosystems like RSA and Elliptic Curve Cryptography (ECC). Understanding groups, subgroups, and mappings between them (homomorphisms) is essential for grasping the security mechanisms of modern encryption protocols.

1.2 Definition and Examples of Groups

1.2.1 Theory

Definition 1.2.1.1 (Group): A **group** is a set G equipped with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the following axioms:

1. **Closure:** For all $a, b \in G$, the result $a \cdot b$ is also in G .
2. **Associativity:** For all $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity Element:** There exists an element $e \in G$ such that for every element $a \in G$, $e \cdot a = a \cdot e = a$.
4. **Inverse Element:** For each $a \in G$, there exists an element $b \in G$, commonly denoted a^{-1} , such that $a \cdot b = b \cdot a = e$.

If the operation is also commutative (i.e., $a \cdot b = b \cdot a$ for all $a, b \in G$), the group is called an **abelian group**.

Example: The set of integers $\mathbb{Z}$ under addition is an infinite abelian group. The identity is 0, and the inverse of a is $-a$.

Example: The set $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ under addition modulo n is a finite abelian group of order n .

1.2.2 Examples

Example: Modular Addition: In $\mathbb{Z}_{26}$ (often used for classical ciphers), computing $15 + 18 \bmod 26$:

$$15 + 18 = 33 \equiv 7 \pmod{26}$$

Example: Cryptography: Groups are the foundation of many cryptographic primitives.

- **Diffie-Hellman Key Exchange** relies on the difficulty of the Discrete Logarithm Problem in a cyclic group.
- **RSA** uses the multiplicative group of integers modulo n , denoted $\mathbb{Z}_n^*$.

1.2.3 Exercises

1. Verify if the set of non-zero real numbers $\mathbb{R}^*$ under multiplication forms a group.

Solution: Yes, $(\mathbb{R}^*, \cdot)$ is a group.

- **Closure:** The product of two non-zero real numbers is non-zero.
- **Associativity:** Multiplication is associative in $\mathbb{R}$.
- **Identity:** $1 \in \mathbb{R}^*$ satisfies $1 \cdot a = a$.
- **Inverse:** For every $a \in \mathbb{R}^*$, $\frac{1}{a} \in \mathbb{R}^*$ is the inverse.

1. Compute the inverse of 3 in the group $\mathbb{Z}_{11}$ under multiplication.

Solution: We need x such that $3x \equiv 1 \pmod{11}$. Testing values: $3(4) = 12 \equiv 1 \pmod{11}$. Thus, $3^{-1} = 4$.

1. Determine if the set of 2×2 matrices with integer entries forms a group under matrix addition.

Solution: Yes. Matrix addition is closed, associative, has an identity (zero matrix), and inverses (negative matrix). Since entries are integers, their sum/negation remains integers.

1.3 Subgroups

1.3.1 Theory

Definition 1.3.1.1 (Subgroup): A subset H of a group G is called a **subgroup** if H itself forms a group under the same operation defined on G . We denote this as $H \leq G$.

Proposition 1.3.1.1 (Subgroup Test): A non-empty subset $H \subseteq G$ is a subgroup if and only if for all $a, b \in H$, $a \cdot b^{-1} \in H$.

Theorem 1.3.1.1 (Lagrange's Theorem): If G is a finite group and H is a subgroup of G , then the order (number of elements) of H divides the order of G . i.e., $|H|$ divides $|G|$.

Proof: The left cosets of H in G form a partition of G . The map $h \mapsto gh$ is a bijection between H and gH , so every coset has the same size $|H|$. If there are k distinct cosets, then $|G| = k |H|$. Thus $|H|$ divides $|G|$. ■

1.3.2 Examples

Example: Finding Subgroups: To find subgroups of $\mathbb{Z}_6$ (under addition): The order of $\mathbb{Z}_6$ is 6. Possible orders for subgroups are divisors of 6: 1, 2, 3, 6.

- Order 1: $\{0\}$ (trivial subgroup)
- Order 2: $\{0, 3\}$ ($3 + 3 = 6 \equiv 0$)
- Order 3: $\{0, 2, 4\}$ ($2 + 2 = 4, 4 + 2 = 6 \equiv 0$)
- Order 6: $\mathbb{Z}_6$ itself.

Example: Small Subgroup Attacks: Subgroups play a crucial role in preventing “small subgroup attacks” in cryptographic protocols. In Diffie-Hellman, if the group order has small factors, an attacker can confine the shared secret to a small subgroup and exhaustively search for it. We often choose groups of prime order or groups where the order has a large prime factor to avoid this.

1.3.3 Exercises

1. List all subgroups of $\mathbb{Z}_{12}$.

Solution: Divisors of 12 are 1, 2, 3, 4, 6, 12. Subgroups correspond to these orders.

- Order 1: $\{0\}$
- Order 2: $\{0, 6\}$
- Order 3: $\{0, 4, 8\}$
- Order 4: $\{0, 3, 6, 9\}$
- Order 6: $\{0, 2, 4, 6, 8, 10\}$
- Order 12: $\mathbb{Z}_{12}$

1. Propose a reason why we might want to use a subgroup of prime order for cryptography.

Solution: Prime order subgroups have no proper non-trivial subgroups. This prevents small subgroup attacks where an attacker forces the computation into a smaller group to solve the discrete logarithm problem more easily.

1. Let $G = \mathbb{Z}_7^*$ (under multiplication). Find the subgroup generated by the element 2.

Solution: $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$. Powers of 2: $2^1 = 2$ $2^2 = 4$ $2^3 = 8 \equiv 1$ So the subgroup is $\{1, 2, 4\}$. Order is 3.

1.4 Homomorphisms

1.4.1 Theory

Definition 1.4.1.1 (Homomorphism): A **homomorphism** is a structure-preserving map between two algebraic structures. Let $(G, \cdot)$ and $(H, *)$ be groups. A function $\varphi : G \rightarrow H$ is a group homomorphism if for all $a, b \in G$:

$$\varphi(a \cdot b) = \varphi(a) * \varphi(b)$$

Proposition 1.4.1.1 (Properties of Homomorphisms):

- $\varphi(e_G) = e_H$ (maps identity to identity)
- $\varphi(a^{-1}) = \varphi(a)^{-1}$ (maps inverses to inverses)

Proof:

1. $\varphi(e_G) = \varphi(e_G \cdot e_G) = \varphi(e_G) * \varphi(e_G)$. Multiplying by $\varphi(e_G)^{-1}$ gives $e_H = \varphi(e_G)$.
2. $e_H = \varphi(e_G) = \varphi(a \cdot a^{-1}) = \varphi(a) * \varphi(a^{-1})$. Thus $\varphi(a^{-1})$ is the inverse of $\varphi(a)$.

■

Kernel 1.4.1.2: The **Kernel** of φ , denoted $\ker(\varphi)$, is the set $\{g \in G \mid \varphi(g) = e_H\}$. It is always a normal subgroup of G .

1.4.2 Examples

Example: Consider $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ defined by $\varphi(x) = x \bmod n$. Check homomorphism property: $\varphi(x + y) = (x + y) \bmod n = (x \bmod n + y \bmod n) \bmod n = \varphi(x) +_n \varphi(y)$. Thus, it is a homomorphism.

Example: Homomorphic Encryption: Homomorphisms are implicit in many reduction steps in cryptography. For example, the security reduction of many protocols relies on mapping a complex problem instance to another structure where relations are preserved. They are also fundamental in constructing “Homomorphic Encryption” schemes, where computations can be performed on encrypted data ($\varphi(\text{enc}(m_1) \cdot \text{enc}(m_2)) = \varphi(\text{enc}(m_1 \cdot m_2))$), allowing operations on the ciphertext to map to operations on the plaintext.

1.4.3 Exercises

1. Let $\varphi : \mathbb{R}^* \rightarrow \mathbb{R}^*$ be defined by $\varphi(x) = |x|$. Is φ a homomorphism under multiplication?

Solution: Check if $\varphi(xy) = \varphi(x)\varphi(y)$. $\varphi(xy) = |xy| = |x||y| = \varphi(x)\varphi(y)$. Yes, it is a homomorphism.

1. Find the kernel of the homomorphism $\varphi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_4$ defined by $\varphi(x) = x \bmod 4$.

Solution: We need $x \in \mathbb{Z}_{12}$ such that $x \bmod 4 = 0$. The elements in $\{0, 1, \dots, 11\}$ divisible by 4 are 0, 4, 8. $\ker(\varphi) = \{0, 4, 8\}$.

1.5 Automorphisms

1.5.1 Theory

Definition 1.5.1.1 (Automorphism): An **automorphism** of a group G is an isomorphism from G to itself. That is, distinct bijection $\varphi : G \rightarrow G$ that is also a homomorphism.

Proposition 1.5.1.1: The set of all automorphisms of G , denoted $\text{Aut}(G)$, forms a group under function composition.

Proof:

1. **Closure:** The composition of two isomorphisms is an isomorphism.
2. **Associativity:** Function composition is associative.
3. **Identity:** The identity map $\text{id}(x) = x$ is an automorphism.
4. **Inverse:** The inverse function of an isomorphism is also an isomorphism.

■

Inner Automorphism 1.5.1.2: For any $g \in G$, the conjugation map $\varphi_{g(x)} = gxg^{-1}$ is an automorphism.

1.5.2 Examples

Example: Let $G = \mathbb{Z}_5$ (additive). Any automorphism must map a generator to a generator. Generators of $\mathbb{Z}_5$ are $\{1, 2, 3, 4\}$. Maps: $\sigma_1(x) = 1x = x$ (Identity)
 $\sigma_2(x) = 2x \bmod 5$ $\sigma_3(x) = 3x \bmod 5$ $\sigma_4(x) = 4x \bmod 5$ Each of these is an automorphism.

Example: Structure Weaknesses: Automorphisms describe the symmetries of the algebraic structure. In cryptography, understanding the automorphism group of the underlying structure (like an elliptic curve) is vital for understanding potential weaknesses or for optimizing arithmetic (e.g., using Frobenius automorphisms for faster scalar multiplication on Koblitz curves).

1.5.3 Exercises

1. Determine all automorphisms of the group $\mathbb{Z}_6$.

Solution: Automorphisms of $\mathbb{Z}_n$ map generators to generators. Generators of $\mathbb{Z}_6$ are elements coprime to 6: 1, 5.

- $\sigma_1(x) = 1x = x$ (Identity)
- $\sigma_5(x) = 5x = -x$

So $\text{Aut}(\mathbb{Z}_6) \simeq \mathbb{Z}_2$.

1. Show that for an abelian group, the only inner automorphism is the identity map.

Solution: Inner automorphism: $\varphi_{g(x)} = gxg^{-1}$. Since G is abelian, $gx = xg$. Thus, $\varphi_{g(x)} = xgg^{-1} = xe = x$. This is the identity map for all g .

1.6 Cayley's Theory

1.6.1 Theory

Theorem 1.6.1.1 (Cayley's Theorem): Every group G is isomorphic to a subgroup of the symmetric group acting on G . In other words, every group can be considered as a group of permutations.

Proposition 1.6.1.1 (Permutation Representation): Specifically, for each $g \in G$, we can define a permutation $\lambda_g : G \rightarrow G$ by $\lambda_{g(x)} = g \cdot x$. The mapping $g \mapsto \lambda_g$ is an injective homomorphism from G into S_G (the group of all permutations of the set G).

Proof:

1. **Well-defined:** λ_g is a permutation because it has an inverse $\lambda_{g^{-1}}$.
2. **Homomorphism:** $\lambda_{gh}(x) = (gh)x = g(hx) = \lambda_{g(\lambda_h(x))}$. Thus $g \mapsto \lambda_g$ preserves operations.
3. **Injective:** If $\lambda_g = \lambda_h$, then $\lambda_{g(e)} = \lambda_{h(e)} \Rightarrow g = h$.

Thus G is isomorphic to the image $\{\lambda_g \mid g \in G\}$, which is a subgroup of S_G .

■

1.6.2 Examples

Example: Consider the group $G = \{1, -1\}$ under multiplication. Elements: $1, -1$. Permutations corresponded to left multiplication: $\lambda_1: 1 \rightarrow 1, -1 \rightarrow -1$ (Identity permutation $(1)(-1)$) $\lambda_{-1}: 1 \rightarrow -1, -1 \rightarrow 1$ (Transposition $(1, -1)$) Thus $G \simeq \{(1)(-1), (1, -1)\} \subseteq S_2$.

Example: Significance: Cayley's theorem is theoretically significant as it allows us to represent abstract groups concretely using permutations. In computational group theory, permutation representations are often easier to store and manipulate in a computer than abstract definitions, allowing for efficient algorithms to determine group properties.

1.6.3 Exercises

1. Write down the permutation representation of $\mathbb{Z}_3 = \{0, 1, 2\}$ given by Cayley's Theorem.

Solution: Elements: 0, 1, 2. Operation is addition mod 3.

- $\lambda_0: 0 \rightarrow 0, 1 \rightarrow 1, 2 \rightarrow 2 \rightarrow (0)(1)(2) = e$
- $\lambda_1: 0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 0 \rightarrow (0\ 1\ 2)$
- $\lambda_2: 0 \rightarrow 2, 1 \rightarrow 0, 2 \rightarrow 1 \rightarrow (0\ 2\ 1)$

1. Explain why the image of the map $g \mapsto \lambda_g$ is a subgroup of S_G .

Solution: The map is a homomorphism. The image of any group homomorphism is a subgroup of the codomain. Since the codomain is S_G , the image is a subgroup of S_G .

1.7 Permutation Groups

1.7.1 Theory

Definition 1.7.1.1 (Permutation Group): A **permutation group** is a group G whose elements are permutations of a given set M and whose group operation is function composition. The symmetric group S_n consists of all permutations of a set of n elements and has order $n!$.

Proposition 1.7.1.1 (Cycle Decomposition): Any permutation can be decomposed into disjoint cycles. The order of a permutation is the least common multiple (LCM) of the lengths of these disjoint cycles.

Proof: Consider the limit of the sequence $x, \sigma(x), \sigma^2(x), \dots$ for any x . Since the set is finite, it must loop back to x . This forms a cycle. Removing these elements and repeating the process decomposes the entire permutation into disjoint cycles. For disjoint cycles, operations commute. Order is the smallest k such that $\text{LCM}(l_1, \dots, l_m)$ divides k . ■

1.7.2 Examples

Example: In S_5 , let $\sigma = (1\ 2\ 3)(4\ 5)$. This is a composition of a 3-cycle and a 2-cycle. Order of $\sigma = \text{lcm}(3, 2) = 6$. Computing σ^2 : $\sigma^2 = (1\ 2\ 3)^2(4\ 5)^2 = (1\ 3\ 2)(e) = (1\ 3\ 2)$.

Example: Ciphers: Permutation groups are the basis of many classical ciphers (like the Enigma machine, which used complex permutations). In modern design, substitution-permutation networks (SPNs) used in block ciphers like AES rely heavily on mixing (permutations) and substitution (S-boxes, which are often defined via other algebraic properties) to achieve confusion and diffusion.

1.7.3 Exercises

1. Calculate the product of permutations $\pi = (1\ 3\ 4)$ and $\sigma = (1\ 2)(3\ 4)$ in S_4 .

Solution: Compute from right to left (standard function composition): $\pi\sigma$.

- $1 \xrightarrow{\sigma} 2 \xrightarrow{\pi} 2$
- $2 \xrightarrow{\sigma} 1 \xrightarrow{\pi} 3$
- $3 \xrightarrow{\sigma} 4 \xrightarrow{\pi} 1$
- $4 \xrightarrow{\sigma} 3 \xrightarrow{\pi} 4$

Result: $(1\ 2\ 3)$.

1. Find the order of the permutation $(1\ 2\ 4)(3\ 5\ 6)$ in S_6 .

Solution: It's a product of disjoint cycles of length 3 and 3. Order = $\text{lcm}(3, 3) = 3$.

1. How many elements does S_4 have? List 5 different elements.

Solution: $|S_4| = 4! = 24$. Examples: $e, (1\ 2), (1\ 3), (1\ 2\ 3), (1\ 2)(3\ 4)$.

2 Ring Theory

2.1 Introduction

Ring theory studies algebraic structures with two binary operations: addition and multiplication. Rings generalize the arithmetic of integers. They are crucial in cryptography because most cryptographic algorithms (like RSA, AES, ECC) operate over specific finite rings or fields (which are special types of rings).

2.2 Definition and Examples of Rings

2.2.1 Theory

Ring 2.2.1.1: A **ring** is a set R equipped with two binary operations, usually denoted by $+$ (addition) and $\cdot$ (multiplication), satisfying:

1. $(R, +)$ is an abelian group (additive identity is 0 , inverse of a is $-a$).
2. $(R, \cdot)$ satisfies associativity $(a \cdot (b \cdot c) = (a \cdot b) \cdot c)$.
3. Distributive Laws hold: $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

Definition 2.2.1.2:

- If there is a multiplicative identity (denoted 1 such that $1 \cdot a = a \cdot 1 = a$), it is a **ring with unity**.
- If multiplication is commutative, it is a **commutative ring**.

Definition 2.2.1.3 (Field): A **field** is a commutative ring with unity where every non-zero element has a multiplicative inverse.

2.2.2 Examples

Example: In $\mathbb{Z}_{12}$, arithmetic is done modulo 12.

$$5 \times 7 = 35 \equiv 11 \pmod{12}$$

Note that $\mathbb{Z}_{12}$ is not a field because not all elements have inverses (e.g., 2 has no inverse).

Example:

- **Finite Fields:** $\text{GF}(p)$ (or $\mathbb{Z}_p$) where p is prime, are fields widely used in ECC.
- **Integers modulo n :** $\mathbb{Z}_n$ is a ring used in RSA.

2.2.3 Exercises

1. Is $\mathbb{Z}$ under component-wise addition and multiplication a ring?

Solution: The phrasing “component-wise” usually applies to vectors. If referring to $\mathbb{Z}$, standard addition and multiplication form a ring. If referring to $\mathbb{Z} \times \mathbb{Z}$, then yes, component-wise operations make it a ring. $\mathbb{Z}$ itself is a ring.

1. Find all units (invertible elements) in the ring $\mathbb{Z}_{10}$.

Solution: Units in $\mathbb{Z}_n$ are elements coprime to n . For $n = 10$, units are $\{1, 3, 7, 9\}$.

2.3 Ideals and Quotient Rings

2.3.1 Theory

Definition 2.3.1.1 (Ideal): An **ideal** I of a ring R is a subring such that for all $r \in R$ and $x \in I$, both $r \cdot x$ and $x \cdot r$ are in I . Ideals “absorb” multiplication by ring elements.

Definition 2.3.1.2 (Quotient Ring): Given an ideal I , we can form the **quotient ring** R/I . The elements of R/I are cosets $a + I$, and operations are defined naturally:

- $(a + I) + (b + I) = (a + b) + I$
- $(a + I) \cdot (b + I) = (a \cdot b) + I$

Proposition 2.3.1.1 (Kernel is an Ideal): The kernel of any ring homomorphism $\varphi : R \rightarrow S$ is an ideal of R .

Proof: Let $K = \ker(\varphi)$.

1. **Subgroup:** K is an additive subgroup (standard group theory result).
2. **Absorption:** Let $r \in R, k \in K$. $\varphi(r \cdot k) = \varphi(r) \cdot \varphi(k) = \varphi(r) \cdot 0 = 0$. $\varphi(k \cdot r) = \varphi(k) \cdot \varphi(r) = 0 \cdot \varphi(r) = 0$. Thus $r \cdot k \in K$ and $k \cdot r \in K$, so K is an ideal.

■

2.3.2 Examples

Example: Consider $R = \mathbb{Z}$ and $I = n\mathbb{Z}$ (multiples of n). I is an ideal. The quotient ring $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$. Calculation in $\mathbb{Z}/5\mathbb{Z}$ is just arithmetic modulo 5.

Example: AES S-Box: Quotient rings are fundamental in constructing finite fields consisting of polynomial bases, e.g., $\text{GF}(2^8) \simeq \mathbb{Z}_2[x]/P(x)$ where $P(x)$ is an irreducible polynomial. This structure is the basis for the AES S-box.

2.3.3 Exercises

1. Show that the set of even integers $2\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Solution: Let $x \in 2\mathbb{Z}$. Then $x = 2k$ for some k . For any $r \in \mathbb{Z}$, $r \cdot x = r(2k) = 2(rk) \in 2\mathbb{Z}$. Thus, it absorbs multiplication and is an ideal.

1. In $\mathbb{Z}_{[x]}$, let $I = \langle x^2 + 1 \rangle$. Describe the elements of $\mathbb{Z}_{[x]}/I$.

Solution: Elements of the quotient are polynomials modulo $x^2 + 1$. Any polynomial $f(x)$ can be written as $q(x)(x^2 + 1) + r(x)$ where $\deg(r) < 2$. So elements are of the form $ax + b$ with $a, b \in \mathbb{Z}$. (This is isomorphic to the Gaussian integers $\mathbb{Z}[i]$).

2.4 Euclidean Rings

2.4.1 Theory

Euclidean Domain 2.4.1.1: A **Euclidean Domain** (or Euclidean Ring) is an integral domain R equipped with a Euclidean value function (or norm) $N : R \setminus \{0\} \rightarrow \mathbb{N}$ allowing a division algorithm. For any $a, b \in R$ with $b \neq 0$, there exist $q, r \in R$ such that:

$$a = bq + r$$

where either $r = 0$ or $N(r) < N(b)$.

2.4.2 Examples

Example: $\mathbb{Z}$ is a Euclidean domain with $N(a) = |a|$. Example: $a = 23, b = 5$. $23 = 5(4) + 3$. Here $q = 4, r = 3$, and $|3| < |5|$.

Theorem 2.4.2.1 (Extended Euclidean Algorithm (EEA)): The Extended Euclidean Algorithm (EEA) finds the greatest common divisor (GCD) and coefficients x, y such that $ax + by = \gcd(a, b)$.

Proof: The algorithm proceeds by successive divisions: $r_{i-2} = q_i r_{i-1} + r_i$. Since $N(r_i)$ is a strictly decreasing sequence of non-negative integers, it must terminate with $r_k = 0$. The last non-zero remainder r_{k-1} divides all previous remainders and linear combinations, so it is the GCD. Back-substitution yields x, y . ■

Example: key Generation:

- **RSA Key Generation:** The EEA is used to compute the modular multiplicative inverse of the public exponent e to find the private key d ($ed \equiv 1 \pmod{\varphi(n)}$).
- **Determining Inverses:** In any finite field, finding inverses is done via the Euclidean algorithm.

2.4.3 Exercises

1. Use the Extended Euclidean Algorithm to find the inverse of 17 modulo 26.

Solution: We want $17x \equiv 1 \pmod{26}$. $26 = 1(17) + 9$ $17 = 1(9) + 8$ $9 = 1(8) + 1$
Back substitute: $1 = 9 - 8$ $1 = 9 - (17 - 9) = 2(9) - 17$ $1 = 2(26 - 17) - 17 = 2(26) - 3(17)$ So $-3(17) \equiv 1 \pmod{26}$. $-3 \equiv 23 \pmod{26}$. Inverse is 23.

1. Prove that the ring of Gaussian Integers $\mathbb{Z}[i]$ is a Euclidean domain.

Solution: Define norm $N(a + bi) = a^2 + b^2$. For any $z, w \in \mathbb{Z}[i]$, consider $\frac{z}{w}$ in $\mathbb{C}$. This lies in a unit square of the lattice $\mathbb{Z}[i]$. The standard Euclidean distance to the nearest lattice point is at most $\frac{\sqrt{2}}{2} < 1$. This implies existence of a remainder with strictly smaller norm.

2.5 Polynomial Rings

2.5.1 Theory

Definition 2.5.1.1 (Polynomial Ring): A **Polynomial Ring** $R[x]$ consists of polynomials with coefficients in a ring R . The division algorithm for polynomials holds if R is a field.

Definition 2.5.1.2 (Irreducible Polynomial): A polynomial $P(x)$ is **irreducible** over F if it cannot be factored into two non-constant polynomials in $F[x]$.

Theorem 2.5.1.1 (Finite Field Construction): Finite fields of order p^n , denoted $\text{GF}(p^n)$, are constructed as $\mathbb{Z}_{p[x]}/\langle P(x) \rangle$ where $P(x)$ is an irreducible polynomial of degree n .

Proof: Let $F = \mathbb{Z}_{p[x]}/\langle P(x) \rangle$. F is a commutative ring with unity. Let $f(x) \in F$ be non-zero. Since $P(x)$ is irreducible and $\deg(f) < \deg(P)$, $\gcd(f, P) = 1$. By EEA, there exist $a(x), b(x)$ such that $a(x)f(x) + b(x)P(x) = 1$. In the quotient ring (mod $P(x)$), this implies $a(x)f(x) \equiv 1$. Thus every non-zero element has an inverse, making F a field. ■

2.5.2 Examples

Example: In $\mathbb{Z}_2[x]$, perform addition and multiplication. Let $f(x) = x^2 + x + 1$ and $g(x) = x + 1$. $f(x) + g(x) = (x^2 + x + 1) + (x + 1) = x^2 + 2x + 2 = x^2$ (coefficients mod 2). $f(x) \cdot g(x) = (x^2 + x + 1)(x + 1) = x^3 + x^2 + x + x^2 + x + 1 = x^3 + 2x^2 + 2x + 1 = x^3 + 1$.

Example: Cryptography:

- **AES:** Operates in $\text{GF}(2^8)$ constructed with the irreducible polynomial $x^8 + x^4 + x^3 + x + 1$.
- **ECC:** Uses fields $\text{GF}(2^m)$ or $\text{GF}(p)$.

2.5.3 Exercises

1. Factor $x^2 - 1$ in $\mathbb{Z}_5[x]$.

Solution: $x^2 - 1 = (x - 1)(x + 1) = (x + 4)(x + 1)$ in $\mathbb{Z}_5$.

1. Determine if $x^3 + x + 1$ is irreducible in $\mathbb{Z}_2[x]$.

Solution: For degree 3, it is reducible iff it has a root. Test elements of $\mathbb{Z}_2$: $0 : 0^3 + 0 + 1 = 1 \neq 0$. $1 : 1^3 + 1 + 1 = 3 \equiv 1 \neq 0$. No roots, so it is irreducible.